

Eric (Ding) Ding

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EDUCATION

Cornell University

Ph.D., Computer Science, Direction: Systems and Networking, GPA: 4.0

Ithaca, NY, United States

Aug. 2024 – June 2029 Expected

University of Michigan

B.S.E., Computer Science, Summa Cum Laude, GPA: 3.96, Dual Degree Program

Ann Arbor, MI, United States

Aug. 2022 – Apr. 2024

Shanghai Jiao Tong University

B.S.E., Electrical and Computer Engineering, Outstanding Graduate, GPA: 3.76, top 10%

Shanghai, China

Sep. 2020 – Aug. 2024

RESEARCH EXPERIENCE

Graduate Student Researcher

Cornell University

Ithaca, New York

Aug. 2024 – Present

- Developing physical layer routing algorithms for silicon photonic fabrics in rack-scale ML networks.
- Building compute and networking resource scheduler for reconfigurable optical networks based on TPU torus topology.

Machine Learning System Research Assistant

SymbioticLab, University of Michigan

Ann Arbor, MI, United States

May 2023 – Apr. 2024

- Developed *Propius*, a Federated Learning (FL) resource management system, based on a microservice architecture.
- Built and deployed a distributed FL evaluation framework in GPU clusters, supporting multi-job parallel training.
- Implemented *Venn*, an advanced scheduling policy, in *Propius* that improves the average FL job completion time by up to 1.88x compared to random allocation.
- Contributed to open-source project *FedScale*, the largest benchmark for FL, in adaptive FL optimizer implementation.

Embedded System Research Assistant

The Fan Lab, University of Michigan

Ann Arbor, MI, United States

May 2023 – Sep. 2023

- Developed *WASP*, a wireless wearable device, capable of monitoring sweat for early disease detection.
- Designed a reliable communication protocol that operates atop I2C and Bluetooth Low Energy protocols, ensuring high-fidelity data communication between two in-device microcontrollers and a terminal microcontroller.

Machine Learning Theory Research Assistant

John Hopcroft Center, Shanghai Jiao Tong University

Shanghai, China

Sep. 2021 – Mar. 2022

- Studied, implemented, and theoretically analyzed Non-Contrastive Self-Supervised Learning (SSL) algorithms.
- Conducted experiments, and compared the performances of Non-Contrastive SSL and traditional Supervised Learning, showing the robustness of SSL methods on imbalanced datasets with long-tail distribution.

WORK EXPERIENCE

Artificial Intelligence Internship

Bosch

Shanghai, China

May 2024 – Aug. 2024

- Designed and implemented a multi-agent system using LangGraph, OpenAI endpoints, and Neo4J graph database for retrieval augmented generation and domain-specific task reasoning.
- Deployed the system on Microsoft Azure for internal use, achieving an estimated 30% reduction in man-hour.

PUBLICATION

- **Eric Ding**, Rachee Singh, PipSwitch: A Circuit Switch Using Programmable Integrated Photonics, Optical Fiber Communication (OFC) Conference 2025 (arXiv:2501.18136)
- Jiachen Liu, Fan Lai, **Ding Ding**, Yiwen Zhang, Mosharaf Chowdhury, Venn: Resource Management Across Federated Learning Jobs, MLSys 2025 (arXiv:2312.08298)
- Anjali Devi Sivakumar, Ruchi Sharma, Chandrakalavathi Thota, **Ding Ding**, Xudong Fan, WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss, ACS Sensors 2023.

VOLUNTEERING AND ACTIVITIES

Lunch and Learn Czar

Computer Science Department, Cornell University

Ithaca, NY, United States

Feb. 2025 - Present

- Facilitate catering and presentation technology for department seminars.

Computer Science Peer Tutor

Renew CS Program, School of Information, University of Michigan

Ann Arbor, MI, United States

Feb. 2024 - May. 2024

- Provided weekly group and individual tutoring sessions for EECS 280 Programming and Data Structures.

Electrical Engineer

Shanghai Jiao Tong University Racing Team

Shanghai, China

Mar. 2021 - Aug. 2022

- Designed a carbon fiber dashboard using Catia. Performed low-voltage electrical system wiring.
- Our team won national second prize of 2021 Formula Student Combustion China.

PROJECTS

A Fully Synthesizable Out-of-Order RISC-V Processor | *SystemVerilog, Verdi*

- Designed an out-of-order processor with features including: RISC-V R10k style register renaming scheme, N-way superscalar, load/store queue, tournament branch predictor, and non-blocking cache.
- Wrote the processor and a complete set of testcases in SystemVerilog, used Verdi platform for debugging and verification.

CNN Convolution Optimization | *CUDA, Slurm*

- Optimized forward convolution layer computation in CUDA, adopting parallel programming techniques, such as memory coalescing, shared memory multiplication and loop unrolling.
- Achieved a processing time of 0.079 seconds for large batches (10K $33 \times 33 \times 12$ images with 24 filters).

Simplified Operating System Kernel | *C++*

- Developed a CPU scheduler and a thread library, supporting thread allocation, interruption, and synchronization primitives, such as mutex and conditional variables.
- Developed a pager which manages processes' virtual address spaces, and swap-backed and file-backed pages across physical memory and disk. Built a multi-threaded network file server in UNIX file system hierarchy.

Distributed Search Engine | *Python, React, AWS*

- Developed a fault-tolerant distributed system running MapReduce framework. Created segmented inverted indexes of web pages through a MapReduce pipeline that is compatible with Hadoop Streaming.
- Implemented a distributed backend index service, capable of generating customized search results via PageRank and TF-IDF integration, and a scalable frontend search server.

GRANTS AND AWARDS

- Graduate School Fellowship, Cornell University, Feb. 2024
- James B. Angell Scholar, University of Michigan, Feb. 2024
- University Honors, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Dean's List, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Tang Junyuan JI Scholarship Nominee, Aug. 2022
- Shanghai Jiao Tong University Merit Student Award, Nov. 2021

TECHNICAL SKILLS

Languages: C++, Python, C, SystemVerilog, Go, RISC-V, Matlab, Javascript, Bash, SQL, R

Frameworks: CUDA, NCCL, PyTorch, Tensorflow, JAX, React, gRPC, Hadoop, LangGraph

Developer Tools: Git, Docker, Kubernetes, Redis, Neo4J, Cloudflare, AWS, Azure, L^AT_EX, VSCode, Arduino, STM32CubeIDE, Wireshark, Gurobi

Simulation and Modeling: Verdi, Catia, Matlab, Mathematica, LabVIEW, Pspice, Proteus, Vivado